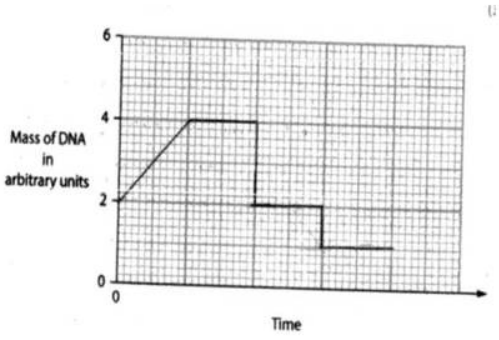
Type MS · Compare DNA with RNA

Answer 1 of 10 · Question 4 part a.iii · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
4	(a) (i)	**A; (chromosomes)** B/C/D do not contain DNA		1
	(ii)	RNA/mRNA/tRNA;		1
	(iii)	<table><tr><th>Differences</th><th>DNA</th><th>Named nucleic acid</th></tr><tr><td>1</td><td>double stranded</td><td>single stranded</td></tr><tr><td>2</td><td>contains deoxyribose</td><td>contains ribose</td></tr><tr><td>3</td><td>contains thymine</td><td>contains uracil</td></tr></table>		3
	(b) (i)	• replication; • formation of a new strand (of DNA); • by complementary base pairing; • double the original mass of DNA;		4
	(ii)	• cell dividing; • cytokinesis;		2
	(iii)	• line starting at two; • increases to four; • ends at one;  		3

Total 14 marks

Type MS · Compare DNA with RNA

Answer 1 of 10 · Question 4 part a.iii · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare DNA with RNA

Answer 2 of 10 · Question 2 part c · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

2	(a) (i)			1
	(ii)	In the following order only: phagocytes; enzymes;		 1 1
	(iii)	produce antibodies/release antitoxins;		1
	(b)	<table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table>	One mark for each correct row	3
	(c)	**D** (a single-stranded helix containing the bases AUGC); A is incorrect as RNA is not a double-stranded helix B is incorrect as RNA is not a double-stranded helix C is incorrect as RNA does not contain the base T		1

Total question 2 = 8 marks

Type MS · Compare DNA with RNA

Answer 2 of 10 · Question 2 part c · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare DNA with RNA

Answer 3 of 10 · Question 2 part b · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number				
2	(a) (i)	• genetic material made of RNA (1) • no DNA (1)		2
	(ii)	<table><tr><th>Disease</th><th>Blood Tested (✓)</th></tr><tr><td>anaemia</td><td></td></tr><tr><td>cystic fibrosis</td><td></td></tr><tr><td>gonorrhoea</td><td>✓</td></tr><tr><td>HIV</td><td>✓</td></tr><tr><td>scurvy</td><td></td></tr></table>	-1 for each extra tick	2
	(iii)	• donor/blood transfused into person with Ebola (1) • needs to be compatible/matched/same group/not rejected (1) • otherwise agglutination/clumping occurs (1)	R clotting	3
	(iv)	• cause blood cells to burst/cells are damaged/destroyed/lose structure (1) • results in loss of function (1)		2
	(v)	• antibodies in donated blood (1) • can destroy virus in infected person (1)		2
	(b)	• uracil present, not thymine (1) • guanine would pair with cytosine/adenine with uracil (1) • percentage of G and C/A and U would be the same (1)		3

Total for question = 14 marks

Type MS · Compare DNA with RNA

Answer 3 of 10 · Question 2 part b · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare DNA with RNA

Answer 4 of 10 · Question 1 part b.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

1 (a) (i)	**D** sugars A/B/C are incorrect as amino acids/bases/protein do not form part of backbone		1
(ii)	**B** bases A is incorrect as the structures labelled Y are bases C is incorrect as proteins are not incorporated into DNA structure D is incorrect as sugar is found in the backbone		1
(iii)	Any two from the following: • hydrogen bonds; • between complementary bases;	Allow description of complementary base pairing	2
(iv)	nucleus;		1
(b) (i)	Any two from the following: • RNA contains ribose sugar and DNA contains deoxyribose; • RNA is single-stranded and DNA is double stranded; • RNA contains uracil and DNA contains thymine;		2
(ii)	**D** mRNA and tRNA A/B/C are incorrect as pRNA/dRNA/dRNA are not types of RNA		1
(c)	**R** phosphate; **S** sugar/deoxyribose; **T** base;	Allow a named base	3
			Total 11

Type MS · Compare DNA with RNA

Answer 4 of 10 · Question 1 part b.i · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare DNA with RNA

Answer 5 of 10 · Question 1 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

1 (a) (i)	**D** sugars A/B/C are incorrect as amino acids/bases/protein do not form part of backbone		1
(ii)	**B** bases A is incorrect as the structures labelled Y are bases C is incorrect as proteins are not incorporated into DNA structure D is incorrect as sugar is found in the backbone		1
(iii)	Any two from the following: • hydrogen bonds; • between complementary bases;	Allow description of complementary base pairing	2
(iv)	nucleus;		1
(b) (i)	Any two from the following: • RNA contains ribose sugar and DNA contains deoxyribose; • RNA is single-stranded and DNA is double stranded; • RNA contains uracil and DNA contains thymine;		2
(ii)	**D** mRNA and tRNA A/B/C are incorrect as pRNA/dRNA/dRNA are not types of RNA		1
(c)	**R** phosphate; **S** sugar/deoxyribose; **T** base;	Allow a named base	3
			Total 11

Type MS · Compare DNA with RNA

Answer 5 of 10 · Question 1 part b.ii · 1 mark

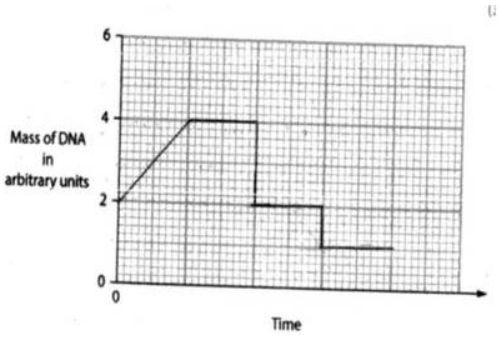
2022 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare DNA with RNA

Answer 6 of 10 · Question 4 part a.ii · 1 mark

2023 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
4	(a)	(i)	**A; (chromosomes)** B/C/D do not contain DNA		1
		(ii)	RNA/mRNA/tRNA;		1
		(iii)	<table><tr><th>Differences</th><th>DNA</th><th>Named nucleic acid</th></tr><tr><td>1</td><td>double stranded</td><td>single stranded</td></tr><tr><td>2</td><td>contains deoxyribose</td><td>contains ribose</td></tr><tr><td>3</td><td>contains thymine</td><td>contains uracil</td></tr></table>		3
	(b)	(i)	• replication; • formation of a new strand (of DNA); • by complementary base pairing; • double the original mass of DNA;		4
		(ii)	• cell dividing; • cytokinesis;		2
		(iii)	• line starting at two; • increases to four; • ends at one;  		3

Total 14 marks

Type MS · Compare DNA with RNA

Answer 6 of 10 · Question 4 part a.ii · 1 mark

2023 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare DNA with RNA

Answer 7 of 10 · Question 2 part a.i · 2 marks

2024 JUN · Paper 02 · find part a.i in this mark scheme extract

Question number				
2	(a) (i)	• nucleus; • cytoplasm/RER/ribosomes;	Allow mitochondrion as an alternative mp	(2)
	(ii)	Any two from: • RNA is single stranded/DNA double stranded; • RNA contains uracil/DNA contains thymine ; • RNA contains ribose /DNA contains deoxyribose;	Allow DNA is double helix Do not allow thiamine	(2)
	(iii)	nucléotide ;	Allow mononucleotides	(1)
	(b)	Any three from • incorrect base/ nucleotide/codon sequence; • incorrect amino acid sequence/amino acid; • faulty/different/incorrect protein made; • reference to protein being a different shape/has a different function;	Allow named mutation/substitution/ deletion/addition/ insertion/eq	(3)

Total for Question 2 = 8 marks

Type MS · Compare DNA with RNA

Answer 8 of 10 · Question 2 part a.ii · 2 marks

2024 JUN · Paper 02 · find part a.ii in this mark scheme extract

Question number				
2	(a) (i)	• nucleus; • cytoplasm/RER/ribosomes;	Allow mitochondrion as an alternative mp	(2)
	(ii)	Any two from: • RNA is single stranded/DNA double stranded; • RNA contains uracil/DNA contains thymine ; • RNA contains ribose /DNA contains deoxyribose;	Allow DNA is double helix Do not allow thiamine	(2)
	(iii)	nucléotide ;	Allow mononucleotides	(1)
	(b)	Any three from • incorrect base/ nucleotide/codon sequence; • incorrect amino acid sequence/amino acid; • faulty/different/incorrect protein made; • reference to protein being a different shape/has a different function;	Allow named mutation/substitution/ deletion/addition/ insertion/eq	(3)

Total for Question 2 = 8 marks

Type MS · Compare DNA with RNA

Answer 9 of 10 · Question 2 part a.ii · 1 mark

2025 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a)	(i)	X = (pentose) sugar/deoxyribose; Y = phosphate;	reject ribose reject phosphorus	2
	(ii)	DNA/deoxyribonucleic acid;		1
	(iii)	• double/two stranded; • thymine present; • only found in DNA (not RNA);		3
	(iv)	nucleus;		1
(b)	(i)	hydrogen bonds;		1
	(ii)	• connect/hold/join, two strands/bases, together; • between complementary bases (pairs)/nucleotides; • adenine thymine/guanine cytosine;	ignore just A T G C	3
				11 marks

Type MS · Compare DNA with RNA

Answer 9 of 10 · Question 2 part a.ii · 1 mark

2025 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare DNA with RNA

Answer 10 of 10 · Question 2 part a.iii · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

2 (a)	(i)	X = (pentose) sugar/deoxyribose; Y = phosphate;	reject ribose reject phosphorus	2
	(ii)	DNA/deoxyribonucleic acid;		1
	(iii)	• double/two stranded; • thymine present; • only found in DNA (not RNA);		3
	(iv)	nucleus;		1
(b)	(i)	hydrogen bonds;		1
	(ii)	• connect/hold/join, two strands/bases, together; • between complementary bases (pairs)/nucleotides; • adenine thymine/guanine cytosine;	ignore just A T G C	3
				11 marks

Type MS · Compare DNA with RNA

Answer 10 of 10 · Question 2 part a.iii · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks